

** Student uniform + student card

** Big and clear handwriting

** Name & ID on A4-note; submit it with exam paper for bonus point

1. Data exploration & preprocessing (chapters 2-3)

- Handling missing data
 - Elimination whole attributes, whole records, pros & cons
 - Imputation mean/mode, prediction (e.g. KNN, regression), pros & cons
- Attribute transformation
 - Nominal to numeric one-hot encoding (k binary attributes for k categories)
 - Numeric to nominal discretization : equal width, equal frequency, by entropy
 - Numeric transform normalization, standardization
- Attribute selection
 - Useless attributes
 - Attribute scoring general concepts (no specific method)
 - Subset selection forward, backward
- Questions
 - Explain concepts, compare between 2 approaches/methods
 - Given a dataset → identify problems, choose preprocessing & explain why or how to do it

2. Decision tree, rule induction (chapter 4)

→ explainable models

- Tree construction (nominal + numeric)
 - Attributes 1 by 1 selection based on Gini, entropy, info gain, gain ratio
 - Splitting binary vs. multi-way split
 - Pruning pre-pruning vs. post-pruning
- Rule induction (nominal + numeric)
 - OneR rule set by only 1 attribute
 - Sequential covering general to specific (RIPPER) vs. specific to general
 - Attributes 1 by 1 selection based on accuracy, coverage
 - Rule set property
 - exhaustive default rule
 - mutual exclusive conflict resolution

3. KNN, Naïve Bayes (chapter 5)

- KNN (nominal + numeric) → no model
 - Distance function Euclidean, nominal, mixed Euclidean
 - Attributes manual preprocessing (type, scale, weight/selection)
 - K (#neighbors) too high, too low, proper K
 - Prediction majority vote, weighted vote

- Naïve Bayes (nominal + numeric)
 - Naïve assumption
 - Prediction
 - Attributes
- probability model
- conditional prob.
relative freq. + Laplace correction (nominal)
normal distribution (numeric)

4. Neural network, SVM (chapter 6)

- Neural network (numeric)
 - Multilayer perceptron
 - Node processing
 - Attributes
 - Learning rate, momentum
 - SVM (numeric)
 - Linear hyperplane
 - Attributes
 - Kernel (dot, polynomial)
 - Multi-class classification
- advanced (nonlinear) models
- normalization, type conversion
- input 1 node per attribute
hidden avg. between #input, #output
output 1 node per class
- summation, activation (Sigmoid)
all considered together by weight adjustment
too high, too low
- normalization, type conversion
binary classification, margin
all considered together by hyperplane function
nonlinear to linear data
1 against 1, 1 against all training

Questions

- Basic calculation → best split (DT), prob. (Naïve Bayes)
- Explain concepts + constraints + classification model of individual methods
- Effects of parameters (too high, too low)

5. Performance (chapter 7)

- Training vs. testing performance
 - Overfitting vs. underfitting
 - Formal measurement
 - Confusion matrix
 - Overall accuracy
 - TP, TN, FP, FN
 - Precision, recall, F-measure
 - Sensitivity, specificity
 - Questions
 - Explain concepts, compare between 2 approaches/methods
 - Given confusion matrices of 2-3 classifiers → calculate and compare performance
- reasons, model characteristics, performance
split validation, cross validation, pros & cons